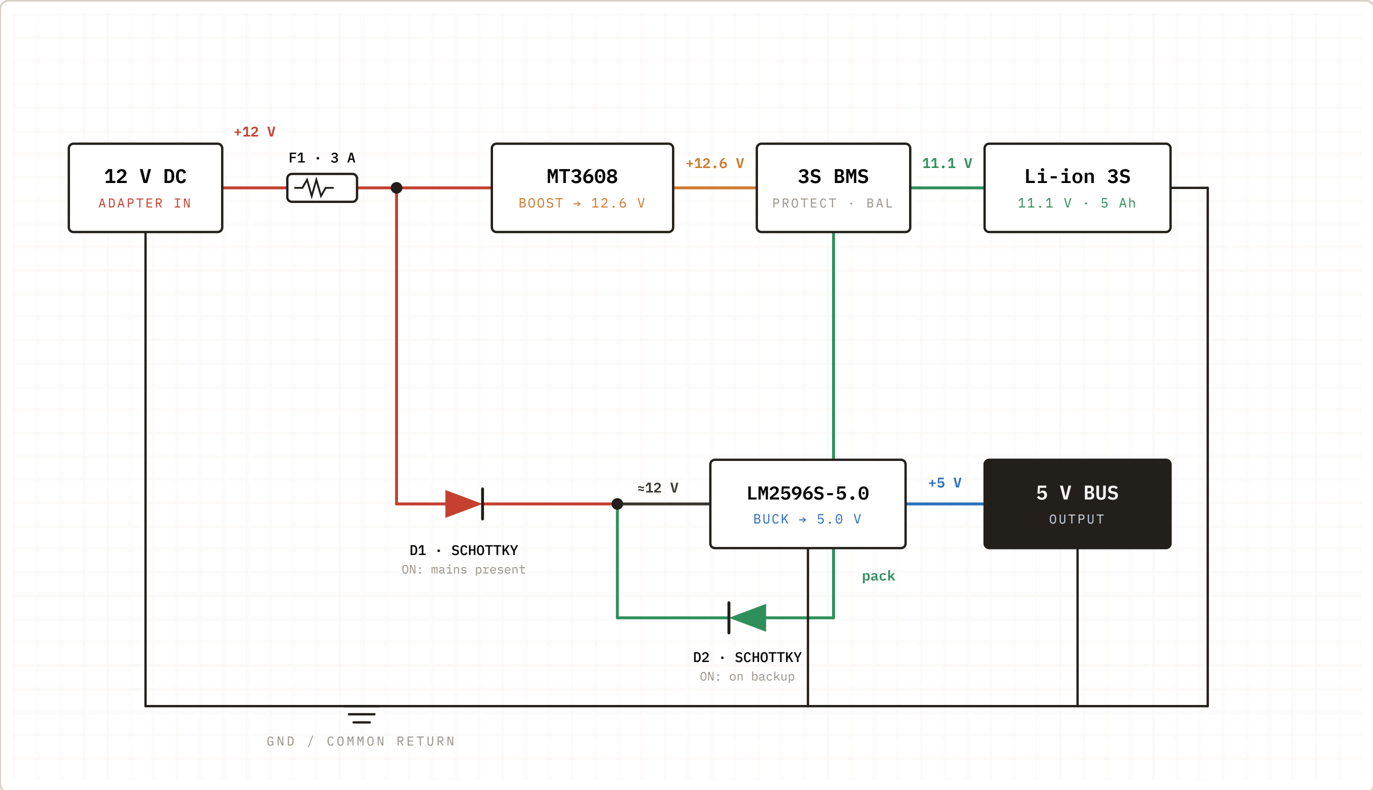
Uninterruptible 5 V Backup Supply

12 V adapter → 3S Li-ion charger + diode-OR 5 V output bus



+12 V input · +12.6 V charge · pack 11.1 V · +5 V output · GND

The adapter performs two jobs in parallel: it charges the 3-cell pack through the MT3608 boost converter and BMS, and it powers the load through diode **D1**. The battery feeds the same output node through diode **D2**. Whichever source sits at the higher voltage wins automatically, so the 5 V bus survives an adapter failure with no interruption — this passive switch is called a **diode-OR**.

1. **12 V DC Adapter (input).** The wall supply. Feeds two branches in parallel — charging the battery and running the load. Its negative ties to the common ground rail.
2. **F1 — 3 A Fuse.** Series input protection. A downstream short or converter fault blows the fuse and isolates the adapter before wiring overheats. Rate it just above the real peak current (charge + load).
3. **Node A — the split.** After the fuse the +12 V rail forks: one branch climbs to the charger, the other drops to the output through D1.
4. **MT3608 — Boost converter → 12.6 V.** A step-up module. A full 3S pack sits at 4.2 V × 3 = 12.6 V, higher than the 12 V input, so the charger must boost slightly above the pack to push current in. Set this voltage on its trim pot *before* connecting the battery.
5. **3S BMS — Battery Management board.** "3S" = manages 3 series cells. It **balances** the cells, enforces **over-charge / over-discharge cutoff** (~12.6 V top, ~9 V bottom), and provides **over-current / short protection**. Charge flows in from the MT3608; discharge flows out to the load through the same terminals.
6. **Li-ion 3S Pack — 11.1 V / 5 Ah.** The energy store. 11.1 V is nominal (3 × 3.7 V); it swings 12.6 V full → ~9 V empty. 5 Ah ≈ 55 Wh — roughly 11 hours at a 0.5 A load, before conversion losses.
7. **D1 — Schottky diode (adapter → bus).** Lets the adapter feed the output but **blocks reverse current**, so the battery can't back-feed the charger. Schottky type for its low forward drop (~0.3 V), wasting little voltage and heat.
8. **D2 — Schottky diode (battery → bus).** Mirror of D1 on the battery side. Lets the pack feed the output but blocks the adapter from forcing current into the pack's output terminal.
9. **Diode-OR node (B).** D1 and D2 meet here and the **higher-voltage source wins automatically**: adapter present → D1 conducts, D2 off; adapter lost → D2 conducts, D1 off. The switchover is instantaneous — no relay, no firmware, no output glitch.
10. **LM2596S-5.0 — Buck converter → 5.0 V.** A step-down regulator with fixed 5.0 V output. It takes the ~9–12 V at the OR node and regulates it to a clean, constant 5 V regardless of the active source.
11. **5 V Bus (output).** The protected rail your load (microcontroller, sensors, etc.) runs on. It never sees the source change.
12. **GND / Common return.** All negatives — adapter, pack, both converters and the load — share one ground. Without this common reference the diode-OR and converters cannot work.

OPERATING MODES

SIGNAL	MAINS POWER	BATTERY BACKUP
Adapter	Supplying	Absent
D1 (adapter)	**Conducting**	Blocked
D2 (battery)	Blocked	**Conducting**
MT3608 charger	Charging pack	Idle
5 V bus source	From adapter	From battery

KEY SPECIFICATIONS

12.6 V
pack full-charge / boost set-point

11.1 V · 5 Ah
nominal pack ≈ 55 Wh

5.0 V
regulated output bus

BUILD & SAFETY NOTES

- Pre-set the MT3608 to 12.6 V with the battery **disconnected**; connecting first can over-volt the pack.
- Use Schottky diodes rated above your peak load current; the small forward drop means D1/D2 still run slightly warm — heatsink if >1 A.
- All grounds must be common — adapter −, BMS B−, both converter GNDs and the 5 V bus return.
- Size F1 just above peak (charge + load) current; add fusing on the battery line for a fully protected pack.
- The BMS only protects the cells — it does not regulate the output; that is the LM2596's job.